

Ritika Kalia

Computer Science & Engineering

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EDUCATION

Bachelor of Engineering in Computer Science & Engineering	2024 – 2027
Chandigarh College of Engineering & Technology	CGPA: 9.45/10 (till 6th sem)
Diploma in Computer Science & Engineering	2022 – 2024
Chandigarh College of Engineering & Technology	86%
Intermediate / Class XII	2020 – 2021
Shishu Niketan Model Sr. Sec. School	91.8%
Matriculation / Class X	2018 – 2019
Shishu Niketan Model Sr. Sec. School	95%

TECHNICAL SKILLS

Programming Languages: Python, C, SQL

Web Development: HTML, CSS, JavaScript, Django, Flask

Machine Learning / Deep Learning: PyTorch, Hugging Face Transformers

Concepts: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems

PROJECTS

Graph Algorithm Visualizer GitHub

JavaScript, HTML, CSS, ES6 Modules

- Built an interactive web application to visualize BFS and DFS graph traversal algorithms step by step.
- Designed a modular frontend separating graph data structures, algorithm logic, and UI rendering using ES6 modules.
- Implemented real-time Queue and Stack visualizations synchronized with algorithm execution.
- Added interactive graph features including edge creation, node drag-and-drop, zoom, pan, and dynamic boundary expansion.

CPU Scheduling Simulator GitHub

Python, Streamlit, Matplotlib, Pandas

- Built an interactive simulator implementing 8 CPU scheduling algorithms including FCFS, SJF, SRTN, Round Robin, LJF, LRTN, and Static/Dynamic Priority.
- Implemented CPU and I/O mixed workloads, preemption, context switches, and aging-based starvation prevention.
- Designed a tick-by-tick scheduling engine with Gantt chart visualization and metrics including waiting time, turnaround time, response time, CPU utilization, and throughput.

Full-Stack E-Commerce Platform – StellerCart GitHub

Python, Django, Bootstrap, HTML, CSS, JavaScript

- Built a full-stack e-commerce web application with authentication, Google sign-in, product listings, cart, checkout, and order history.
- Designed 7 related database models using Django ORM for products, carts, orders, delivery options, addresses, reviews, and contact messages.
- Implemented AJAX-based cart updates with CSRF protection and server-side order calculations.
- Integrated Razorpay in test mode with server-side payment amount calculation and payment signature verification.

EXPERIENCE

CS Soft Solutions (I) Pvt. Ltd. Jul 2023 – Aug 2023

Python Programming – Industrial Training

- Completed industrial training focused on Python programming fundamentals and applied scripting.

RESEARCH AND PUBLICATIONS

IEEE ICIR 2026 – Research Publication and e-Poster Presentation

Real-Time Privacy Guardianship in XR Metaverses via AI-Driven Behavioral Analytics

- Developed a behavioral analytics pipeline for detecting VR privacy attacks, achieving 92% accuracy and 0.976 ROC-AUC.
- Presented the research as an e-poster at the University of Pisa, Italy.
- IEEE Xplore GitHub

Publication – Memory in Conversational AI Agents: The Backbone of Long-Term Intelligence

Publication Link

LEADERSHIP AND ACHIEVEMENTS

- **Chairperson**, ACM-W CCET Student Chapter, 2025.
- **Technical Team Member**, CCET APRATIM 2025 College Tech Fest.
- **Design Contributor**, CCET College Website Team; designed official departmental web pages.
- **3rd Position**, Business Blueprint 2025, IIC CCET; pitched a multimodal AI/ML Women Safety Analytics System.